

Variance analysis

- **Direct Material Variance**

- ***Price and Quantity Variances***

- A **price variance** is the difference between the actual price of an input and its standard price, multiplied by the actual amount of the input purchased.
- A **quantity variance** is the difference between how much of an input was actually used and how much should have been used and is stated in dollar terms using the standard price of the input.

EXHIBIT 11-3

A General Model for Variance Analysis—Variable Manufacturing Costs

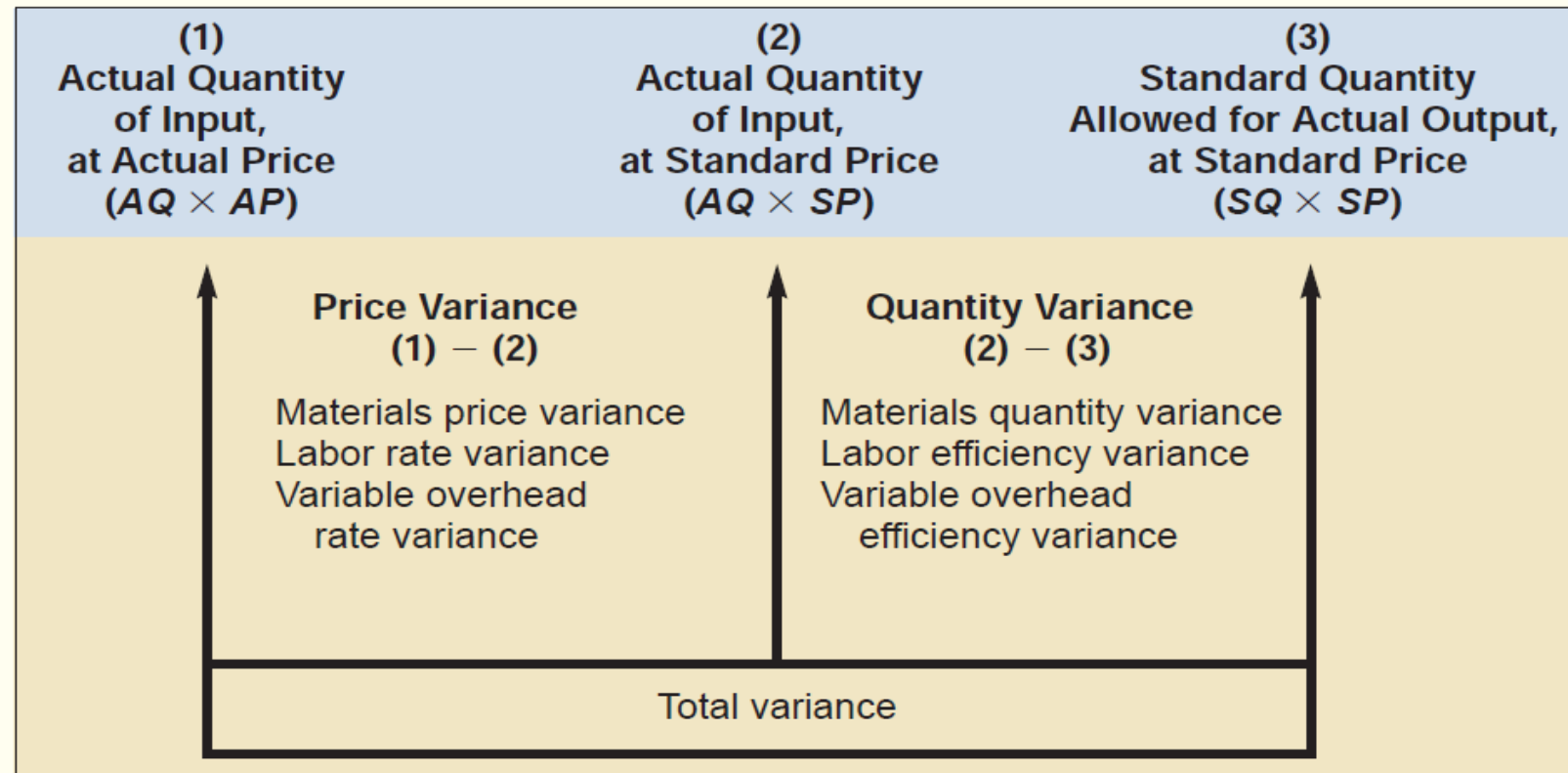


EXHIBIT 11-4

Variance Analysis—Direct Materials

(1) Actual Quantity of Input, at Actual Price ($AQ \times AP$)	(2) Actual Quantity of Input, at Standard Price ($AQ \times SP$)	(3) Standard Quantity Allowed for Actual Output, at Standard Price ($SQ \times SP$)
6,500 pounds $\times$ \$3.80 per pound = \$24,700	6,500 pounds $\times$ \$4.00 per pound = \$26,000	6,000 pounds* $\times$ \$4.00 per pound = \$24,000
Price variance, \$1,300 F		Quantity variance, \$2,000 U
Total variance, \$700 U		

*2,000 units $\times$ 3.0 pounds per unit = 6,000 pounds.
F = Favorable; U = Unfavorable.

Materials price variance = $(AQ \times AP) - (AQ \times SP)$

Actual quantity Actual price Standard price

The formula can be factored as follows:

$$\text{Materials price variance} = AQ(AP - SP)$$

Colonial Pewter Company Variance Report—Purchasing Department						
	(1)	(2)	(3)	(4)		
Item Purchased	Quantity Purchased	Actual Price	Standard Price	Difference in Price (2) – (3)	Total Price Variance (1) × (4)	Explanation
Pewter	6,500 pounds	\$3.80	\$4.00	\$0.20	\$1,300 F	Bargained for an especially good price.
F = Favorable; U = Unfavorable						

- **Direct Labour Variance**

EXHIBIT 11-6

Variance Analysis—Direct Labor

Actual Hours of Input, at the Actual Rate ($AH \times AR$)	Actual Hours of Input, at the Standard Rate ($AH \times SR$)	Standard Hours Allowed for Actual Output, at the Standard Rate ($SH \times SR$)
5,400 hours $\times$ \$13.75 per hour = \$74,250	5,400 hours $\times$ \$14.00 per hour = \$75,600	5,000 hours* $\times$ \$14.00 per hour = \$70,000
Rate variance, \$1,350 F		Efficiency variance, \$5,600 U
Total variance, \$4,250 U		

*2,000 units $\times$ 2.5 hours per unit = 5,000 hours.
F = Favorable; U = Unfavorable.

Labor rate variance.

$$\text{Labor rate variance} = (AH \times AR) - (AH \times SR)$$

Actual hours Actual rate Standard rate

The formula can be factored as follows:

$$\text{Labor rate variance} = AH(AR - SR)$$

EXAMPLE

$$\text{Labor rate variance} = 5,400 \text{ hours } (\$13.75 \text{ per hour} - \$14.00 \text{ per hour}) = \$1,350 \text{ F}$$

- labor efficiency variance

$$\text{Labor efficiency variance} = (AH \times SR) - (SH \times SR)$$

↑ Actual hours
 ↙ ↘ Standard rate Standard hours allowed for actual output

The formula can be factored as follows:

$$\text{Labor efficiency variance} = SR(AH - SH)$$

$$SH = 2,000 \text{ units} \times 2.5 \text{ hours per unit} = 5,000 \text{ hours.}$$

$$\text{Labor efficiency variance} = \$14.00 \text{ per hour} (5,400 \text{ hours} - 5,000 \text{ hours}) = \$5,600 \text{ U}$$

.....

Manufacturing Overhead Variances

EXHIBIT 11-7

Variance Analysis—Variable Manufacturing Overhead

Actual Hours of Input, at the Actual Rate ($AH \times AR$)	Actual Hours of Input, at the Standard Rate ($AH \times SR$)	Standard Hours Allowed for Actual Output, at the Standard Rate ($SH \times SR$)
\$15,390	5,400 hours $\times$ \$3.00 per hour = \$16,200	5,000 hours* $\times$ \$3.00 per hour = \$15,000
	Rate variance, \$810 F	Efficiency variance, \$1,200 U
	Total variance, \$390 U	

*2,000 units $\times$ 2.5 hours per unit = 5,000 hours.
F = Favorable; U = Unfavorable.

- **variable overhead rate variance**

$$\text{Variable overhead rate variance} = (AH \times AR) - (AH \times SR)$$

Actual hours Actual rate Standard rate

This formula can be factored as follows:

$$\text{Variable overhead rate variance} = AH(AR - SR)$$

$$\begin{aligned}\text{Variable overhead rate variance} &= 5,400 \text{ hours } (\$2.85 \text{ per hour} - \$3.00 \text{ per hour}) \\ &= \$810 \text{ F}\end{aligned}$$

- **variable overhead efficiency variance**

$$\text{Variable overhead efficiency variance} = (AH \times SR) - (SH \times SR)$$

The diagram shows the formula $\text{Variable overhead efficiency variance} = (AH \times SR) - (SH \times SR)$ on a yellow background. Below the formula, three labels are positioned: 'Actual hours' under 'AH', 'Standard rate' under the first 'SR', and 'Standard hours allowed for actual output' under 'SH'. Three pink arrows point from these labels to their respective variables in the formula: one from 'Actual hours' to 'AH', one from 'Standard rate' to the first 'SR', and one from 'Standard hours allowed for actual output' to 'SH'. A blue circle highlights the minus sign between the two terms in the formula.

This formula can be factored as follows:

$$\text{Variable overhead efficiency variance} = SR(AH - SH)$$

$$\begin{aligned}\text{Variable overhead efficiency variance} &= \$3.00 \text{ per hour } (5,400 \text{ hours} - 5,000 \text{ hours}) \\ &= \$1,200 \text{ U}\end{aligned}$$

Xavier Company produces a single product. Variable manufacturing overhead is applied to products on the basis of direct labor-hours. The standard costs for one unit of product are as follows:

Direct material: 6 ounces at \$0.50 per ounce	\$ 3
Direct labor: 1.8 hours at \$10 per hour	18
Variable manufacturing overhead: 1.8 hours at \$5 per hour	<u>9</u>
Total standard variable cost per unit	<u>\$30</u>

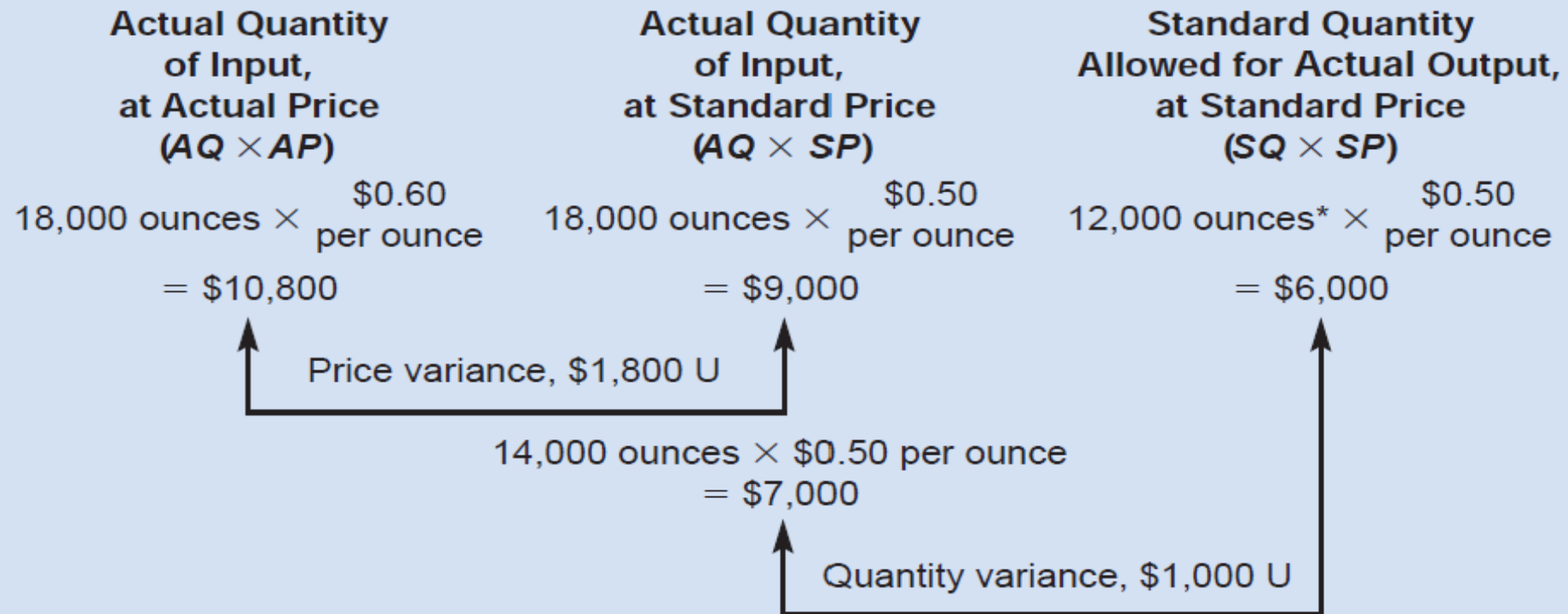
During June, 2,000 units were produced. The costs associated with June's operations were as follows:

Material purchased: 18,000 ounces at \$0.60 per ounce	\$10,800
Material used in production: 14,000 ounces	—
Direct labor: 4,000 hours at \$9.75 per hour	\$39,000
Variable manufacturing overhead costs incurred	\$20,800

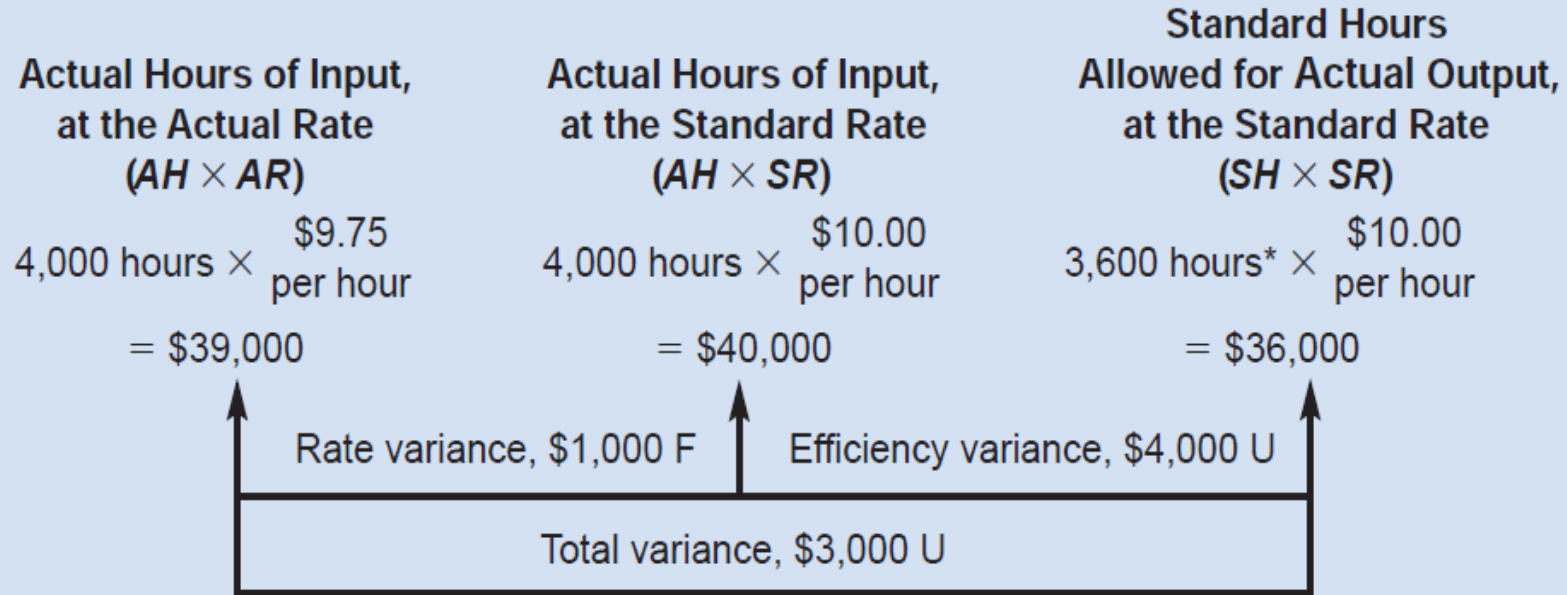
Required:

Compute the direct materials, direct labor, and variable manufacturing overhead variances.

Direct Materials Variances



Direct Labor Variances



Variable Manufacturing Overhead Variances

**Actual Hours of Input,
at the Actual Rate
($AH \times AR$)**

\$20,800

**Actual Hours of Input,
at the Standard Rate
($AH \times SR$)**

4,000 hours $\times$ \$5.00
per hour
= \$20,000

**Standard Hours
Allowed for Actual Output,
at the Standard Rate
($SH \times SR$)**

3,600 hours* $\times$ \$5.00
per hour
= \$18,000

Rate variance, \$800 U

Efficiency variance, \$2,000 U

Total variance, \$2,800 U